Lukas Dimitroff

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Availability: Jan – Jul 2027

EDUCATION

Northeastern University – Khoury College **May 2028**
Bachelor of Science in Computer Science; 3.5/4.0 GPA *Boston, MA*
- Relevant Classes: Mathematical Foundations of AI, Algorithms and Data, and Computer Systems

Norwin High School **Jun 2025**
National Honor Society; 3.955 unweighted / 4.123 weighted GPA *Irwin, PA*

Pennsylvania Governor's School of Science (PGSS) **Jun 2024 – Jul 2024**
Student *Carnegie Mellon University*
- Selected as 1 of 72 from 403 applicants for an intensive program covering CS, discrete math, and AI/ML
- Researched sensor augmentation for OpenFace occluded-face detection to increase detection confidence

TECHNICAL PROJECTS

Arm NEON Kernel Optimization **Aug 2026 – Present**
Independent Developer (PR in progress), ggml / llama.cpp
- Identified that the current implementation was compute-bound due to a missing batched GEMM kernel for IQ4_NL
- Hand-wrote GEMM kernels using Arm NEON SIMD intrinsics to maximize Cortex-A53 performance
- Concluded implemented changes sped up GEMM by 6.8x in isolation and 1.6x end-to-end
- Performing additional tests based on documented requirements for effective communication with PR reviewers

unsloth-turboquant Fork **Jul 2026 – Aug 2026**
Fork Maintainer
- Exposed TurboQuant quantizations in Unsloth Studio, shrinking KV cache by up to 6x
- Built an architecture-aware beellama.cpp installer, ensuring system-specific preprocessor directives are compiled

SEND – Hackonomics Hackathon **Mar 2026**
1st Place, 1,463 participants
- Developed a node-based platform for building and backtesting trading strategies from composable graphs
- Iterated on a backtesting replay view over 6 months of data with daily P/L
- Composed a React frontend, Java/Spring REST API, and OCaml execution engine communicating over stdin/stdout
- Exposed service from homelab through Cloudflare Tunnel, gaining managed TLS, DDoS mitigation, and IP masking

EXPERIENCE

Vast.ai **Jul 2026 – Present**
Infrastructure Host *Irwin, PA*
- Operating an RTX 5090 rental host: Ubuntu 24.04 Server, CUDA 13.0, nvidia-container-toolkit
- Isolated renter containers via iptables/nftables egress rules and UniFi VLAN segmentation
- Investigating kernel panics and segmentation faults likely due to CPU degradation from faulty Intel microcode

Relight Candles (Nonprofit) **Dec 2022 – Aug 2025**
Founder *Irwin, PA*
- Raised and donated over $14,000 to St. Jude by selling handmade, non-carcinogenic soy candles
- Registered the nonprofit, filed tax returns, and tracked inventory counts to ensure 501(c)(3) tax exemption status

Norwin's Castle Production Lab (Not-for-profit) **Sep 2022 – May 2025**
Founder / Director of Finance and Operations *Irwin, PA*
- Composed a webstore and workflows giving special-needs students real work experience
- Presented financial performance and recommendations to the school board that shaped the future of the business

McDonald's **Feb 2021 – Jan 2023**
Crew Member *Irwin, PA*
- Hired at age 14 and given crew lead responsibilities at 15, training new employees and promoting effective teamwork

SKILLS & ACTIVITIES

Languages: Python; C++; Java; C; OCaml; Lua
Frameworks and Tools: ggml; NumPy; Git; nginx; Cloudflare Tunnels; Docker; Arm NEON; UniFi ecosystem
Awards & Activities: Pittsburgh Regional Science & Engineering Fair, Excellence of Presentation; Eurochallenge semifinalist; AFA CyberCamp at Pitt; Westinghouse Honors Science Institute